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HR Expert



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Problem Statement

Problem Statement

Manual HR screening often poses significant challenges such as time-consuming resume reviews, inconsistent evaluation criteria, and difficulty in tracking candidate progress. These issues lead to delays in shortlisting suitable applicants and inefficiencies in the recruitment process. The primary goal of this project is to automate the end-to-end evaluation workflow—from resume submission to HR decision-making—ensuring faster, more consistent, and objective candidate assessments.

Objectives

- Automate the end-to-end HR screening process using AI and workflow automation tools.
- Extract and structure candidate data automatically from uploaded resumes.
- Compare candidate qualifications against predefined job profiles to ensure fair and consistent evaluation.
- Generate AI-based scoring and reasoning for transparency and faster HR decision-making.
- Store all evaluation results in Google Sheets for easy tracking and reporting.
- Send automatic email notifications to candidates regarding their application status.



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Project Overview

Project Overview

- **Accepts candidate CVs through a form:**

Candidates can easily upload their resumes via a web form. The system securely collects and stores submissions for further analysis.

- **Extracts personal and professional data using AI:**

The uploaded PDF resumes are automatically scanned using an AI-powered extraction model that identifies key details such as name, email, contact number, education, skills, and experience.

- **Compares with predefined job profiles:**

The extracted candidate data is matched against predefined job role criteria (e.g., required skills, location, or qualifications). The system evaluates how closely a candidate's profile aligns with the job description.

- **Generates evaluation score and feedback:**

Based on the comparison, an AI module generates a quantitative score and qualitative feedback highlighting strengths, weaknesses, and fit for the role.

- **Logs results in Google Sheets:**

Every candidate's information, evaluation score, and decision are automatically recorded in Google Sheets for easy tracking, reporting, and reference by HR teams.

- **Sends automatic emails to candidates:**

Personalized emails are sent to candidates with tailored messages — whether selected or rejected — ensuring professional and consistent communication.



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Tools and Technologies

Tools and Technologies

Tools/API	Purpose
n8n	Workflow automation & orchestration
OpenAI GPT-4 Turbo	AI-based data extraction & evaluation
Google Drive API	Secure CV storage
Google Sheets API	Logging and analytics
Gmail Integration	Candidate notifications
PDF Extractor Node	Converts CVs into readable text

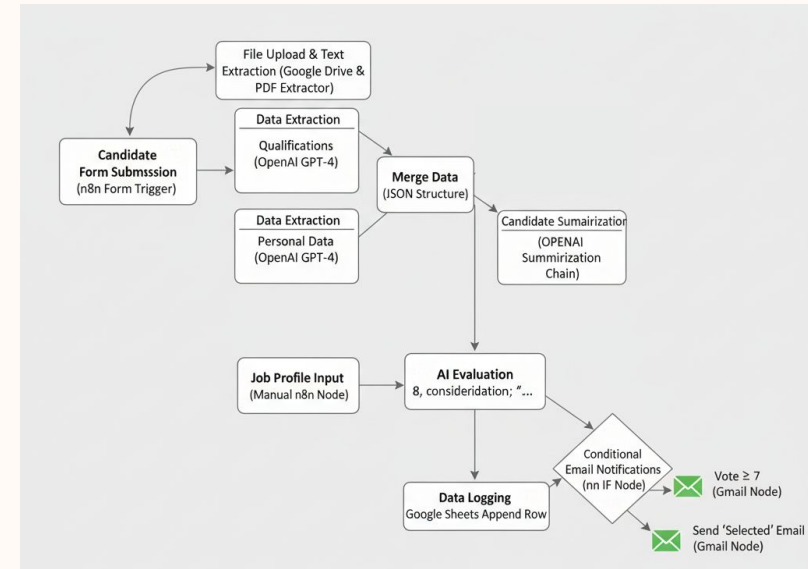


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Workflow Architecture

Workflow Architecture

- Candidate submits form (name, email, CV PDF)
- File uploaded to Google Drive
- AI extracts data (personal + qualifications)
- Merge Node combines both datasets
- Summarization node creates a short candidate summary
- HR Expert model compares it with job profile
- Outputs vote (1–10) and consideration (evaluation text)
- Conditional emails are sent based on score
- Final record stored in Google Sheets





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Node Breakdown

Node Breakdown

Step	Node	Description
1	Form Trigger	Receives candidate submission
2	Upload CV (Drive)	Stores CV in Google Drive
3	PDF Extractor	Extracts text from resume
4	Personal Data (Model)	Extracts name, email, phone
5	Qualifications (Model)	Extracts skills, education, experience
6	Merge Node	Combines both data streams
7	Summarization Chain	Generates profile summary
8	Profile Wanted	HR-defined job requirements
9	HR Expert Model	Evaluates candidate fit
10	If + Email Nodes	Sends result emails
11	Google Sheets	Logs all results

Node Breakdown

Personal / HR Agent + Add tag

0 / 2

Active ☒

Share

Save



Star

153,182

Editor

Executions

Evaluations

HR Agent

This workflow automates the process of handling job applications by extracting relevant information from submitted CVs, analyzing the candidate's qualifications against a predefined profile, and storing the results in a Google Sheet.

The CV is uploaded to Google Drive and converted so that it can be processed

Upload CV

upload: file

Extract from File

Extract from PDF

The essential information for evaluating the candidate is collected in two different chains

Personal Data

Qualifications

Merge

combine

Summary of relevant information useful for classifying the candidate

Summarization Chain

Characteristics of the profile sought by the company that intends to hire the candidate

Profile Wanted

Candidate evaluation with vote and considerations of the HR agent relating the profile sought with the candidate's skills

HR Expert

Logs the results in an excel sheet

Google Sheets

append/Update: sheet

Send Emails to the Candidate based on the voting generated.

Acceptance Mail

send: message

Rejection Mail

send: message

String to Number Code

Switch

write: Rules

On form submission

OpenAI

Structured Output Parser

Execute workflow



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Output

Application Update Inbox

Summarize this email



devarapallihars... 9:53 AM



to me ▾

Hi PAVITRA JASTI, Thank you for your interest in the position. We have carefully reviewed your application and appreciate the time you took to apply. After thorough consideration, we've decided to move forward with other candidates whose experience more closely matches our current needs. Please don't be discouraged — we encourage you to apply again for future opportunities that align with your skills. Wishing you the very best in your career journey. Kind regards, HR Team Recruitment Department

—
This email was sent automatically with [n8n](#)

Suggestions

Thanks for update

Hi, Thanks for the update Best...

Acknowledge reje

Hi, I appreciate yo

Reply

Forward



99+





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Future Enhancements

Future Enhancements

- **Integrate with Airtable / Notion dashboards:** Airtable/Notion give a richer, visual, and collaborative place for HR to review candidates, add notes, and build dashboards — better than raw Sheets for some teams.
- **Enable multi-role evaluation in one workflow:** Evaluate a single candidate against multiple job profiles (e.g., Frontend, Fullstack, Data Analyst) and return role-specific scores.
- **Add ranking and visualization dashboards:** Make results actionable via visual dashboards: top candidates, score distributions, time-to-hire, etc.
- **Incorporate interview scheduling automation:** Automatically schedule interviews for shortlisted candidates (vote $\geq$ threshold), removing back-and-forth emailing.



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Conclusion

Conclusion

HR Expert transforms manual resume evaluation into a smart, automated, AI-assisted process.

It improves accuracy, saves time, and delivers immediate communication to candidates — making HR operations more efficient and scalable.



Thanks!